

Transfer Learning Framework for PM2.5 Forecasting in Indian Cities

A BTP Report

by

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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the BTP entitled “**Transfer Learning Framework for PM2.5 Forecasting in Indian Cities**” in the partial fulfillment of the requirements for the award of the degree of B. Tech and submitted in the Indian Institute of Information Technology SriCity, is an authentic record of my own work carried out during the time period from January 2024 to December 2024 under the supervision of **Dr. Mainak Thakur**, Indian Institute of Information Technology SriCity, India. The matter presented in this report has not been submitted by me for the award of any other degree of this or any other institute.

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ABSTRACT

India is one of the most polluted countries on the planet. PM 2.5 is a silent killer in Indian cities. However, many Indian towns do not have enough monitoring stations to monitor air quality. Our work, titled "Transfer learning framework for PM 2.5 forecasting for Indian cities," tries to solve this problem of forecasting PM 2.5 values in cities with low monitoring stations. In Indian cities with low monitoring stations, data scarcity often arises. In such scenarios, the transfer learning approach is an effective tool. Basically, in Transfer Learning, a pre-trained model, trained on one task is applied to a second related (but not the same) task. This is how the concerns regarding the distribution mismatch can be addressed. In our work, we focused on three cities of India: Delhi, Guwahati, and Kolkata. These cities have 39, 4, and 10 PM 2.5 monitoring stations, respectively. Both in Kolkata and Guwahati less station availability causes scarcity of required data. To solve this problem, we first train the LSTM model to predict PM 2.5 values for each city, which is trained and tested on its own data. Then to fine-tune the model we train it with 15, 30, and 45,60 percent of another city's data respectively, and test it with that city's test data. After that, we train the model with multiple city's data combined and test on the test data of each city (In similar proportions as before). Using this method we get more accurate results for our forecasting.

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1. Introduction

Due to the rapid increase in urbanization in developing countries like India, it is always a challenge to handle pollution. India was the third most polluted country in the world in 2023, with an average PM_{2.5} concentration of 54.4 $\mu\text{g}/\text{m}^3$. Pollutants like PM_{2.5}, PM₁₀ are highly hazardous, impacting human respiratory and cardiovascular systems, causing skin oxidative stress. Additionally, it harms the local ecosystem. Furthermore, it also damages the ecosystem of the local area. To monitor these pollutants there are wide spread of monitoring stations present in India for example cities, towns and rural areas to measure the air quality in different kinds of environment. However, the heterogeneous geography and unique environmental conditions in the northeastern states make it challenging to take care of the spatial variation of PM_{2.5}. Northeast India also lacks the labeled data available in cities like Delhi or Kolkata, hence it is complex to develop accurate models which can be used to forecast PM_{2.5} pollution. There are around 38 monitoring stations of CPCB[\[13\]](#) that are present in Delhi as compared to only 10 in Kolkata and just 4 in Guwahati. The dynamics of air quality in Delhi, Kolkata and Guwahati differ from one region to another due to the climatic, geographical and source of the emission variations. For example, Delhi witnesses extreme levels of PM_{2.5} in winter, contributed by vehicle emissions and crop stubble burning in the surrounding states. Kolkata's air quality is affected by industrial activities and by the proximity of the Hooghly River. Although Less studied, the air pollution in Guwahati is also influenced by its unique topography. Topography and localized sources such as biomass burning and emissions from cars. So doing this study addresses data sparsity as a problem and, through such a mechanism, offers a viable practical alternative for areas that do not enjoy much monitoring infrastructure. This will improve accuracy in the forecasting but

also further enlighten on the adaptability of machine learning models across diverse geographical contexts. So, We use the transfer learning-based approach, which helps to solve the above problem and overcome the data availability issue.

Transfer learning is the machine learning technique in which the model trained using one dataset is used for training on another similar new dataset, rather than building the model on new dataset from scratch. In the present study, we develop a PM_{2.5} forecasting model for Guwahati using LSTM networks: a specialized form of recurrent neural networks known for the ability to catch temporal dependencies in sequential data. Firstly, we trained a base model on the data obtained from Delhi and Kolkata in order to familiarize the model with the diversities of the air quality data and wider network of monitoring stations across these cities; then, the trained model is further fine-tuned for the city of Guwahati to get adapted in the local setting. The model considers multiple features influencing PM_{2.5} levels, including: Ambient Temperature, Relative Humidity, Wind Speed, Wind Direction, Hour of the Day, Month of the Year.

2. Literature Survey

Using Gross Primary Production Data and Deep Transfer Learning for Crop Yield Prediction

[1] This study focuses on leveraging MODIS-derived Gross Primary Production (GPP) data in combination with deep learning models to predict corn and soybean yields in the U.S. Corn Belt. The researchers addressed a critical gap in agricultural applications, where previous models either relied heavily on crop-specific assumptions or lacked robust data transferability. By employing GPP data synthesized from remote sensing sources, the study aimed to build predictive models that could generalize across crop types and locations. The methodology began with the collection of 15 years of crop yield data (2006–2020) from USDA-NASS and the extraction of county-level GPP values for corn and soybean crops. Three deep learning architectures were employed: a Deep Neural Network (DNN) with eight layers, a 1D-Convolutional Neural Network (1D-CNN) for time-series processing, and a Gated Recurrent Unit (GRU) for capturing sequential dependencies in crop growth.

Transfer learning was central to this study, implemented in two key ways: crop-type transfer, where knowledge from corn models was applied to soybean models (and vice versa), and location transfer, which tested the adaptability of models across climatic zones such as the Eastern Temperate Forest and Great Plains. Performance evaluation, conducted using metrics like R^2 and RMSE, revealed that 1D-CNN consistently outperformed the other models, especially when a larger portion of the target dataset (70%) was used for fine-tuning. The results demonstrated the potential of integrating GPP data with deep transfer learning to improve crop yield predictions while reducing computational overhead by reusing pre-trained models.

Transfer Learning in Environmental Data-Driven Models: Ozone Forecast in the Alpine Region

[2] This paper investigates the transferability of machine learning models for forecasting ozone concentrations across diverse environmental settings in the Alpine Arc. The study highlights the challenges posed by data scarcity at many monitoring sites and the importance of transferring knowledge from data-rich locations. Using Long Short-Term Memory (LSTM) neural networks, the researchers aimed to predict hourly ozone concentrations for the next 24 hours based on the previous 48 hours' data. The dataset, spanning from 2007 to 2021, included 20 monitoring stations across urban, rural, and mountainous regions in four European countries. After preprocessing, which involved addressing missing values, the data was split into training, validation, and testing subsets.

The core of the methodology involved developing predictive models at individual sites and assessing their transferability to other locations through cross-testing, without additional retraining. A transferability matrix was constructed to evaluate performance variations across sites, with clustering techniques applied to group similar locations based on environmental and predictive behaviors. The findings revealed that urban and rural models demonstrated higher transferability among similar sites, while mountainous locations posed greater challenges due to their distinct and less variable ozone concentration profiles. By employing metrics like R^2 and Mean Absolute Error (MAE), the researchers quantified the transferability and highlighted the importance of clustering in reducing variability in predictions. Overall, the study provided valuable insights into applying transfer learning for air quality management, emphasizing the potential for refining model adaptability across diverse environmental contexts.

Novel Transfer Learning Framework for Time Series Forecasting

[3] This study introduces TrEnOS-ELMK, a novel framework combining transfer learning, Online Sequential Extreme Learning Machines with Kernels (OS-ELMK), and ensemble learning for time series forecasting. The research addresses the challenges of utilizing time-varying data, particularly when training and testing datasets do not share the same distribution or when fresh data samples are scarce. Traditional time series prediction models struggle in such scenarios due to their reliance on static, abundant datasets. The proposed framework overcomes these limitations by retaining knowledge from older datasets and adapting it for current predictive tasks.

The TrEnOS-ELMK framework is implemented in two phases. In the first phase, knowledge is extracted from older data using OS-ELMK models and transferred to new datasets through adaptive kernel and weight matrix updates. This process ensures the utilization of latent knowledge while accounting for time variability in datasets. In the second phase, ensemble learning is introduced to enhance prediction accuracy. Multiple models are constructed and updated dynamically based on their performance with new incoming data. Poor-performing models are replaced, and adaptive weights are assigned to prioritize high-performing models. The system ensures optimal prediction by maintaining a flexible and evolving ensemble structure.

Experimental validation using synthetic and real-world datasets, such as the Mackey-Glass and Lorenz systems, confirmed the framework's effectiveness. TrEnOS-ELMK demonstrated superior performance compared to traditional models, highlighting its robustness in addressing real-time forecasting challenges with limited fresh data. The integration of transfer learning and ensemble learning marks a significant advancement in time series prediction methodologies.

3. Problem Statement and Contribution

Problem Statement

Air pollution, especially high levels of PM_{2.5}, is a major concern for public health and the environment in Indian cities. Predicting PM_{2.5} levels accurately is essential for timely action, but several challenges exist:

- **Limited Data Availability:** Many cities lack sufficient and continuous air quality data.
- **City-Specific Factors:** Differences in climate, geography, and local activities make it difficult to use one model for multiple cities.
- **Incomplete Data:** Missing or irregular data affects the quality of predictions.
- **High Resource Requirements:** Creating separate models for each city requires significant time and computational effort.

This study aims to overcome these challenges by using transfer learning, which allows knowledge from one city to improve predictions in others, even with limited data.

Contribution

This work makes the following contributions:

1. **Development of a Source Model**

- A Long Short-Term Memory (LSTM) model was trained on Kolkata's air quality data to understand and predict PM2.5 levels.

2. **Transfer Learning Framework**

- The trained model was tested on target cities (Guwahati and Delhi) and fine-tuned with a small amount of their data to improve accuracy.

3. **Improved Data Preparation**

- Advanced methods like interpolation (IDW) and Kalman filtering were used to handle missing data, along with new features like wind direction components and time-based encoding.

4. **Efficient Use of Limited Data**

- The approach demonstrated accurate PM2.5 forecasting in cities with limited data, reducing the need for large datasets or city-specific models.

4. Methodology

Study Area

The study focuses on PM2.5 forecasting in three major Indian cities: Kolkata, Guwahati, and Delhi. These cities represent diverse climatic conditions and air pollution patterns, providing a robust dataset for transfer learning experiments.

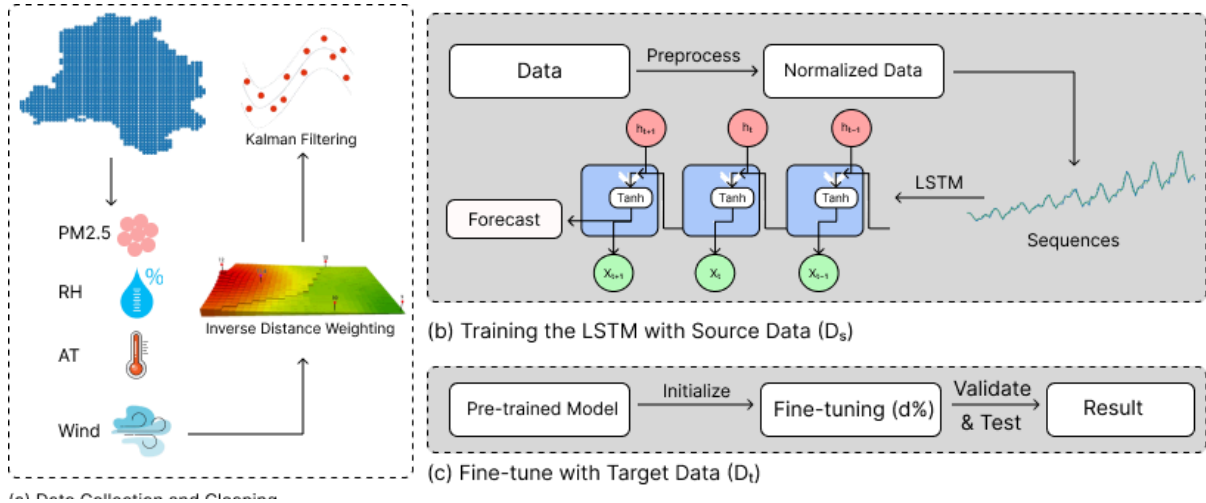


Figure 4.1: Overview of the methodology stages.

4.1 Data Collection

The data used in this study was collected from the Central Pollution Control Board (CPCB[13]) across varying time periods:

- **Kolkata & Guwahati:** January 2022 to December 2023

- **Delhi:** January 2021 to December 2022

Data was recorded at a **3-hour interval**, capturing the following variables:

- **PM2.5:** Fine particulate matter concentration ($\mu\text{g}/\text{m}^3$)
- **Relative Humidity (RH):** Percentage of moisture in the air
- **Air Temperature (AT):** Ambient temperature ($^{\circ}\text{C}$)
- **Wind Speed (WS):** Speed of wind (m/s)
- **Wind Direction (WD):** Direction from which the wind originates (degrees)

4.2 Preprocessing

To ensure consistency and usability for model training, the data underwent the following preprocessing steps:

1. Handling Missing Data

Data gaps were addressed using a combination of:

- **Inverse Distance Weighting (IDW):** For spatial interpolation to estimate missing values based on nearby observations.
- **Kalman Filtering:** For temporal smoothing and filling data gaps, leveraging the sequential nature of air quality measurements.

2. Feature Engineering

Additional features were derived to enhance model performance:

- **Wind Direction Components:** The sine and cosine transformations were computed to represent wind direction in Cartesian coordinates.
- **Time Features:**
 - **Hour of the Day:** Encoded as a one-hot vector to capture diurnal variations.
 - **Month of the Year:** Encoded as a one-hot vector to incorporate seasonal trends.

3. Normalization

All input features were normalized using **StandardScaler** to ensure uniformity and facilitate faster model convergence during training.

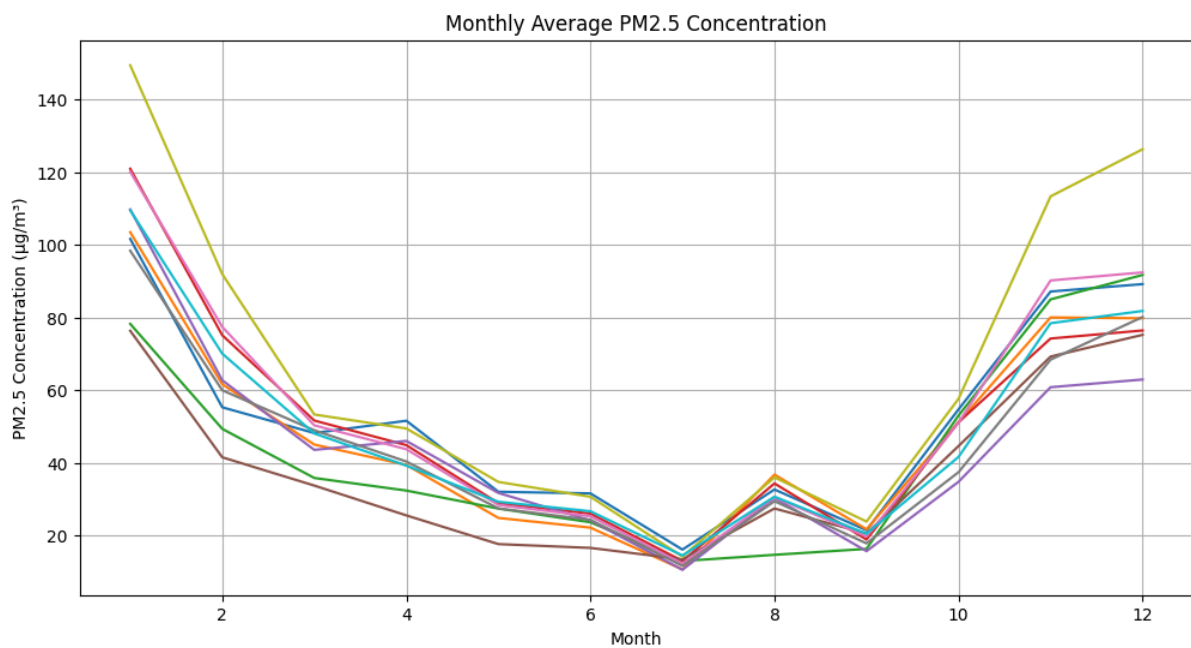


Figure 4.2: Monthly Average PM2.5 Concentration in Kolkata

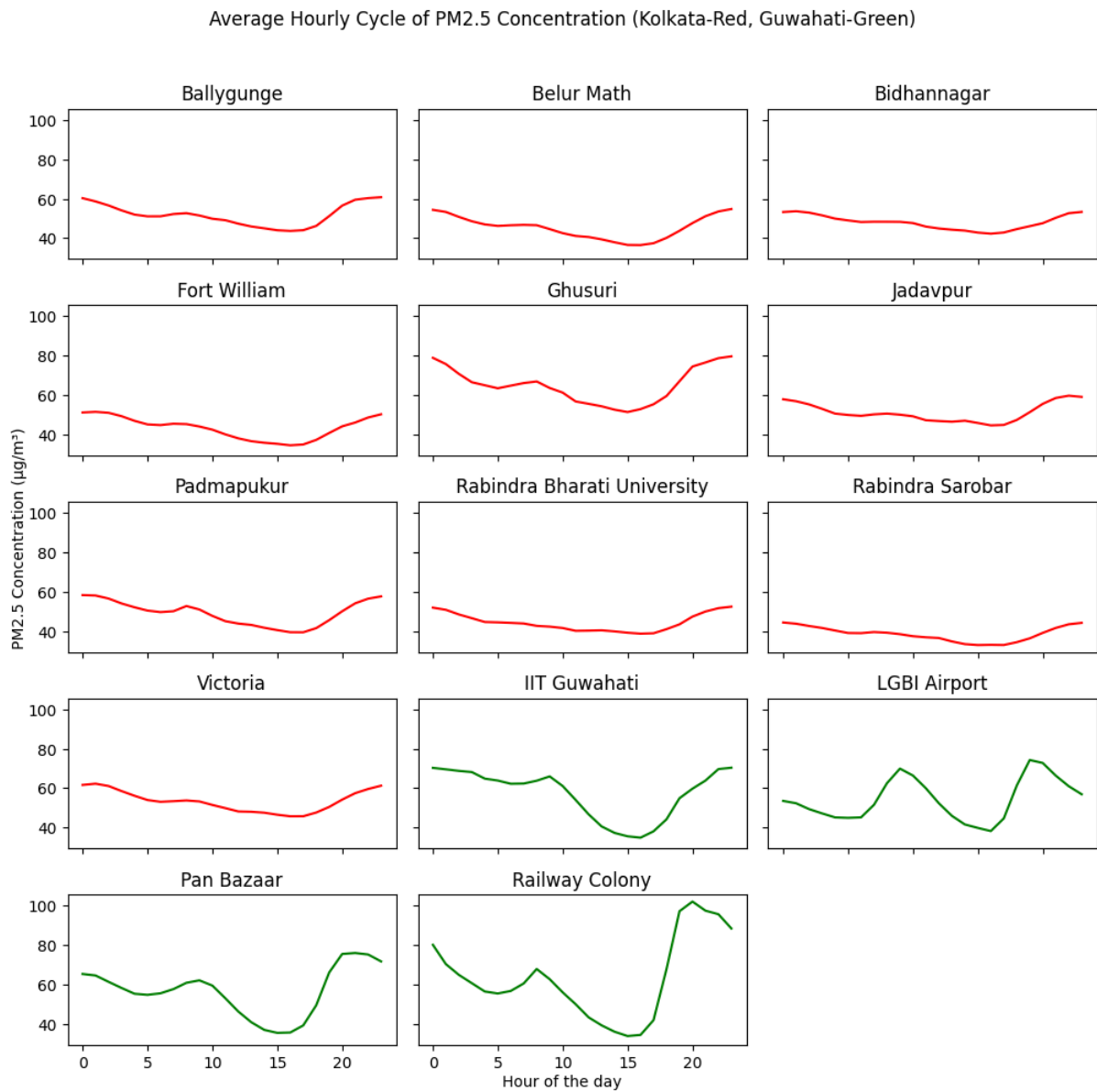


Figure 4.2: Hourly Cycle of Average PM_{2.5} Concentration at Stations in Kolkata & Guwahati

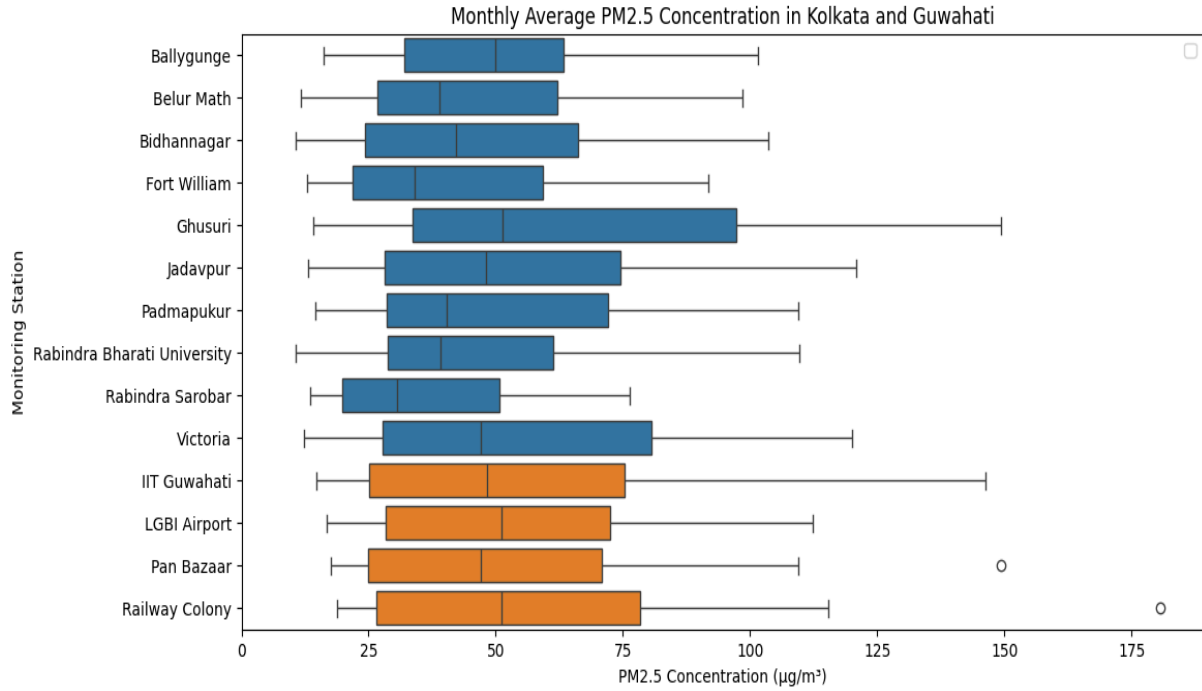


Figure 4.3: Boxplot for Monthly Average Concentration of PM2.5

4.3 Source Model Training

The Long Short-Term Memory (LSTM) model was chosen as the base architecture for PM2.5 forecasting due to its ability to capture long-term temporal dependencies in sequential data.

- The LSTM model was trained on data from **Kolkata**, the **source city**, using a sequence-to-sequence framework.
- **Input:** Historical data over multiple time steps (e.g., 24 hours).
- **Output:** PM2.5 predictions for the subsequent time steps (e.g., next 3 hours).
- The model architecture included multiple LSTM layers with dropout for regularization and fully connected layers for output prediction.
- Optimization was performed using the Adam optimizer with Mean Squared Error (MSE) as the loss function.

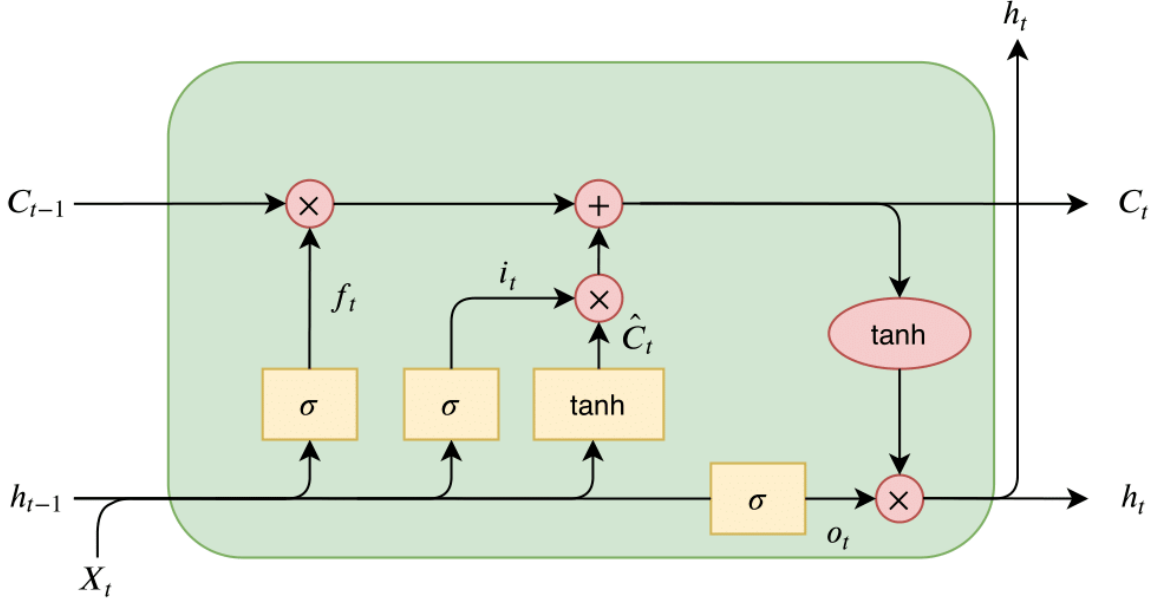


Figure 4.4: Architecture of LSTM

4.4 Transfer Learning Framework

1. Transfer Setup

The trained LSTM model (M_k) was used as the base model for transfer learning experiments in the **target cities** (Guwahati and Delhi). Two scenarios were explored:

- **Direct Testing:** The source model (M_k) was evaluated directly on target city data to assess its generalizability.
- **Fine-tuning:** The source model was fine-tuned using a small fraction ($d\%$) of the target city data.

2. Fine-Tuning Process

Fine-tuning involved:

- Initializing the target city model with pre-trained weights from M_k .
- Training the model on $d\%$ of target city data with a reduced learning rate to retain knowledge from the source city while adapting to target-specific characteristics.
- Exploring multiple values of $d\%$ to analyze the trade-off between data availability and model performance.

5. Results

Performance on Standard Data

Model	R ² score	MAE
M _K	0.7861	14.6633
M _G	0.5723	16.3012
M _D	0.657	37.3301

Table 5.1: Source Model Results (R² Score and MAE)

The model was tested for PM_{2.5} forecasting in three cities: Kolkata (M_K), Guwahati (M_G), and Delhi (M_D). In each case, the city was treated as the source city, and the model was trained and evaluated using its respective dataset. The performance was measured using the R² score, which shows how well the model explains variations in PM_{2.5} levels (higher values indicate better performance), and Mean Absolute Error (MAE), which represents the average prediction error (lower values indicate better accuracy).

For Kolkata (M_K), the model achieved a R² score of 0.7861, indicating strong accuracy in capturing PM_{2.5} variations. The MAE was 14.6633 $\mu\text{g}/\text{m}^3$, reflecting a small prediction error and reliable performance for this city.

In the case of Guwahati (M_G), the R² score was 0.5723, showing moderate accuracy in forecasting PM_{2.5} levels. The MAE was 16.3012 $\mu\text{g}/\text{m}^3$, indicating slightly higher prediction errors compared to Kolkata but still reasonable performance.

For Delhi (M_D), the model achieved a R² score of 0.657, demonstrating its ability to explain a fair amount of the variation in PM_{2.5} levels. However, the MAE was 37.3301 $\mu\text{g}/\text{m}^3$, which reflects higher prediction errors, likely due to the unique and complex pollution patterns in Delhi.

These results suggest that the model performs well when trained individually for each city, with variations in accuracy depending on the local air quality dynamics.

Source	Target	R-Square Score	15%	30%	45%	60%	MAE
Kolkata	Kolkata	0.7861					14.6633
	Guwahati	0.522	0.598	0.603	0.5823	0.6271	15.1105
	Delhi	-0.005	0.6939	0.6908	0.6881	0.687	35.7143
Guwahati	Guwahati	0.5723					16.3012
	Kolkata	0.7211	0.8098	0.8174	0.8164	0.8019	13.5344
	Delhi	0.4946	0.6995	0.6877	0.6815	0.6837	35.91
Delhi	Delhi	0.657					37.3301
	Kolkata	0.7437	0.8189	0.8129	0.8051	0.8069	13.3222
	Guwahati	0.5797	0.6381	0.621	0.6133	0.634	14.5957

Table 5.2: Transfer Learning results

for different combinations using %d amount of target data to finetune the source model

Transfer Learning Analysis

The transfer learning experiments evaluated the effectiveness of using a source city's trained model to forecast PM2.5 levels in a target city, with varying percentages of target city data (15%, 30%, 45%, and 60%) used for fine-tuning. Performance was measured using the **R² score** (higher indicates better prediction capability) and **Mean Absolute Error (MAE)** (lower indicates smaller prediction errors). The key observations are:

Kolkata as the Source City

When Kolkata served as the source city, the model's performance varied significantly across the target cities:

- **Guwahati as Target:**

The R^2 score improved steadily with an increase in fine-tuning data, reaching a maximum of **0.6271** at 60%. The MAE reduced to **15.1105 $\mu\text{g}/\text{m}^3$** , indicating improved accuracy compared to direct testing.

- **Delhi as Target:**

The R^2 score showed a significant jump from **-0.005** (direct testing) to **0.6939** with 15% fine-tuning data, stabilizing near **0.687** as the amount of fine-tuning data increased. The MAE improved to **35.7143 $\mu\text{g}/\text{m}^3$** with fine-tuning, although it remained higher compared to Guwahati.

Guwahati as the Source City

When the model was trained on Guwahati's data, the transfer learning performance for the target cities was as follows:

- **Kolkata as Target:**

Fine-tuning led to a consistent improvement in the R^2 score, peaking at **0.8174** with 45% target data. The MAE dropped to **13.5344 $\mu\text{g}/\text{m}^3$** , demonstrating excellent accuracy for PM2.5 forecasting.

- **Delhi as Target:**

The R^2 score improved from **0.4946** (direct testing) to **0.6995** with 15% fine-tuning data. However, the improvement plateaued beyond 15%, with slight variations. The MAE was **35.91 $\mu\text{g}/\text{m}^3$** , similar to the Kolkata-source scenario for Delhi.

Delhi as the Source City

Using Delhi as the source city yielded the following results for the target cities:

- **Kolkata as Target:**

The model achieved a high R^2 score of **0.8189** with 30% fine-tuning data and remained stable at around **0.8069** as more data was added. The MAE improved to **13.3222 $\mu\text{g}/\text{m}^3$** , showing excellent forecasting performance.

- **Guwahati as Target:**

The R^2 score increased from **0.5797** (direct testing) to **0.6381** with 30% fine-tuning

data, though further increases in fine-tuning data showed only marginal improvement. The MAE was reduced to **14.5957 $\mu\text{g}/\text{m}^3$** , reflecting good accuracy.

6. Conclusion

This study presented a transfer learning framework for PM_{2.5} forecasting in Indian cities. The results showed that the framework could effectively use data from one city to improve air quality predictions in another, even with limited data from the target city.

Fine-tuning the model with a small percentage of target city data (15%-30%) significantly improved its accuracy. The model performed well across different cities, with better results when the target city had similar air quality patterns to the source city. However, cities with complex pollution dynamics, like Delhi, showed higher prediction errors.

The proposed method reduces the need for large datasets and city-specific models, making it a practical solution for cities with limited air quality data. This framework can help policymakers and researchers monitor and manage air pollution more effectively, especially in data-scarce regions.

Future work can focus on improving model performance for cities with complex pollution patterns and exploring the use of additional features to further enhance accuracy.

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List of Abbreviations and Symbols

PM_{2.5} Particulate Matter less than 2.5 micrometers

CPCB Central Pollution Control Board

R² R-squared score

MAE Mean Absolute Error

LSTM Long Short-Term Memory

IDW Inverse Distance Weighting

AT Air Temperature

RH Relative Humidity

WS Wind Speed

WD Wind Direction

$\sim u$ and $\sim v$ Sine and Cosine components of Wind Direction

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